

AKANKSHA TYAGI

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M.S. graduate in Computer Science with experience building AI-integrated web applications using React/Next.js, FastAPI, PostgreSQL, RAG, vector search, and MCP tooling. Motivated to learn, ship reliable features, and collaborate across teams.

EDUCATION

UNIVERSITY OF ARKANSAS **GPA: 3.75/4.0** Fayetteville, AR
Master of Science in Computer Science May 2026

Selected Coursework: Image Processing, Privacy Enhancing Technology, Full Stack Deep Learning, Machine Learning.

COLLEGE OF ENGINEERING ROORKEE (COER) Roorkee, India
Bachelor of Technology in Information Technology May 2017

SKILLS

Languages & Web: Python, TypeScript, JavaScript, SQL, C/C++; React, Next.js, Node.js, FastAPI, REST APIs, HTML/CSS

AI/ML: LLM Integration, RAG, AI Agents, MCP, PyTorch, Reinforcement Learning, Graph Neural Networks

Data & Cloud: PostgreSQL, pgvector, MongoDB, Supabase, AWS, Docker, Git, Linux

PROJECTS

AdvisorDesk — AI-Powered Advisory Content Platform

React · Next.js · TypeScript · FastAPI · Supabase (Postgres + pgvector) · OpenAI · MCP · Docker · AWS

A platform for financial advisory firms where advisors publish guidance through a CMS and clients receive instant, cited answers from an AI assistant grounded in the firm's approved content.

- Built the full stack: two Next.js frontends (an internal CMS for creating, editing, tagging, and publishing advisory content, plus a client app) on a shared FastAPI backend with Google OAuth, containerized with Docker and deployed on AWS.
- Implemented a RAG assistant that answers client questions over published content using vector search (Supabase pgvector) and OpenAI, returning grounded answers with source citations.
- Engineered an agentic content layer that exposes CMS operations as MCP tools, so a content manager can draft, tag, search, and publish through natural language instead of manual UI steps.

Real-time Pedestrian Detection and Safety Alert System (UArk)

Jan 2025 – Apr 2025

- Built a real-time pedestrian detection system on the NVIDIA Jetson Nano using YOLO and MobileNet SSD models trained on the CityPersons and Caltech datasets.
- Optimized edge inference with quantization, pruning, TensorRT, and DeepStream, and developed audio-visual alerts for pedestrians entering restricted roadways and vehicles speeding through red lights.

PROFESSIONAL EXPERIENCE & INTERNSHIPS

Cybersecurity Lab (UArk Affiliated)

Jan 2024 – Jun 2026

Research Assistantship

Advisor: Dr. Qinghua Li

- **Reinforcement Learning for Fuzzing:** Developed reinforcement-learning models for test-case generation to improve vulnerability discovery and code coverage compared with baseline fuzzing methods.
- **Master's Thesis - AI-Based Anomaly Detection in Cyber-Physical Water Testbeds:** Built machine learning models for real-time anomaly detection in water systems, enhancing security and resilience of cyber-physical infrastructure.
- **Generative Vision - Diffusion & LDMs:** Built DDPM and latent-diffusion models for high-fidelity image synthesis and text-to-image generation (cross-attention U-Net), evaluated on ALOT, CelebA-HQ, and LAION with FID/IS.

Reinforcement Learning for Traffic Optimization (IIIT-H Affiliated)

Jun 2022 – Dec 2023

Research Internship

Advisor: Dr. Praveen Paruchuri

- Simulated multimodal traffic environments in SUMO to optimize lane-level dynamics and vehicle-to-vehicle coordination.
- Developed and evaluated reinforcement-learning strategies on grid-world and real-world-inspired road networks, reducing electric-vehicle traversal time compared with baseline methods.

Codeventure Tech Llp

Roorkee, India

Software Engineer

Jun 2021 - Apr 2022

- Designed and developed a city complaint management system using PHP, WordPress, HTML, CSS, and JavaScript, enhancing civic issue reporting and response times.

PUBLICATIONS

- **Tyagi, A., Lowalekar, M. and Paruchuri, P., 2024.** Improving Lane Level Dynamics for EV Traversal: A Reinforcement Learning Approach. In VEHITS (pp. 134-143). DOI: [10.5220/0012637200003702](https://doi.org/10.5220/0012637200003702)
Honors: Nominee for Best Student Paper Award at the VEHITS-2024 conference.
- **Uwibambe, M. L., Tyagi, A., and Li, Q., 2025.** A Reinforcement Learning Approach to Multi-Parametric Input Mutation for Fuzzing. IEEE CSR, pp. 174–179. DOI: [10.1109/CSR64739.2025.11129986](https://doi.org/10.1109/CSR64739.2025.11129986)